

Question 1 — Trustworthy prediction and root-cause identification

A telecom network operator provides you with historical data containing KPIs such as latency, packet loss, throughput, jitter, signal strength, handover failures, CPU/memory utilization, alarms, and configuration changes.

Design a computational-intelligence framework that can:

1. predict network degradation before a serious service disruption occurs;
2. identify the probable root causes of the degradation;
3. distinguish between correlation and causal evidence;
4. provide an explanation that a network engineer can understand; and
5. quantify the uncertainty/confidence of its prediction.

Question: What AI/ML methods would you choose for each component, what would be the input/output of each component, and how would you validate that the framework is trustworthy rather than simply accurate?

What this tests:

- Machine/deep learning knowledge
- Time-series modelling
- Explainable AI
- Uncertainty quantification
- Causal reasoning
- Model validation
- Understanding of telecom networks
- Ability to move from *prediction* to *diagnosis*
- Trustworthy AI
- Practical telecom-management understanding

Find one public data

Write one code to show